Parallel and Distributed Computing
PDC

Ultimate Goal

Faster, more effective programs by exploiting parallelism

What is parallelism?

“The term **Parallelism** refers to techniques to make programs faster by performing several computations <u>at the same time</u>. This requires *hardware with multiple processing units*. ”

https://wiki.haskell.org/Parallelism_vs._Concurrency

Sequential vs. Parallel Programs

Sequential

Each instruction of whole program is executed serially, one at a time

One processing unit

Parallel

At points in a program, execution of some instructions happens concurrently in time

Multiple processing units

We often start with sequential versions of programs and parallelize them to make them faster.

This enables us to solve larger problems.
We are interested in SCALE and scalability of our parallel implementations.

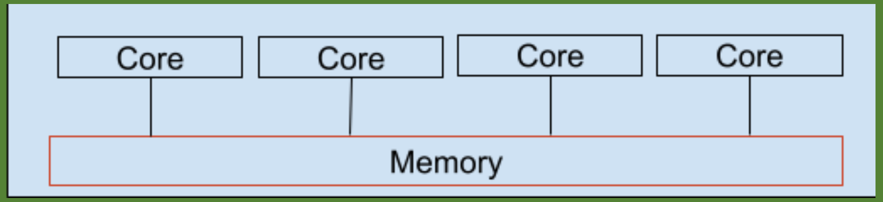
What enables us to create parallelism?

Parallel and Distributed Hardware with multiple *processing units (PUs)*

Software to run on such hardware

Processing Unit Computation: Threads vs. Processes

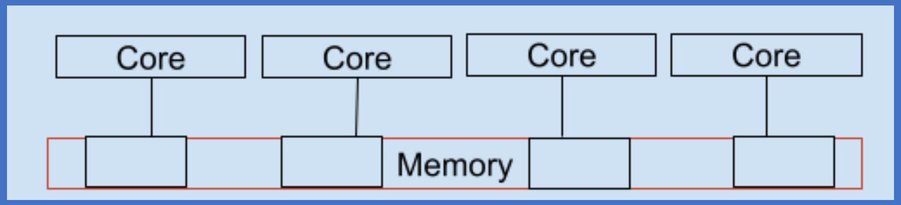
Threads
Share 1 Memory Space



Single shared memory, multicore machine

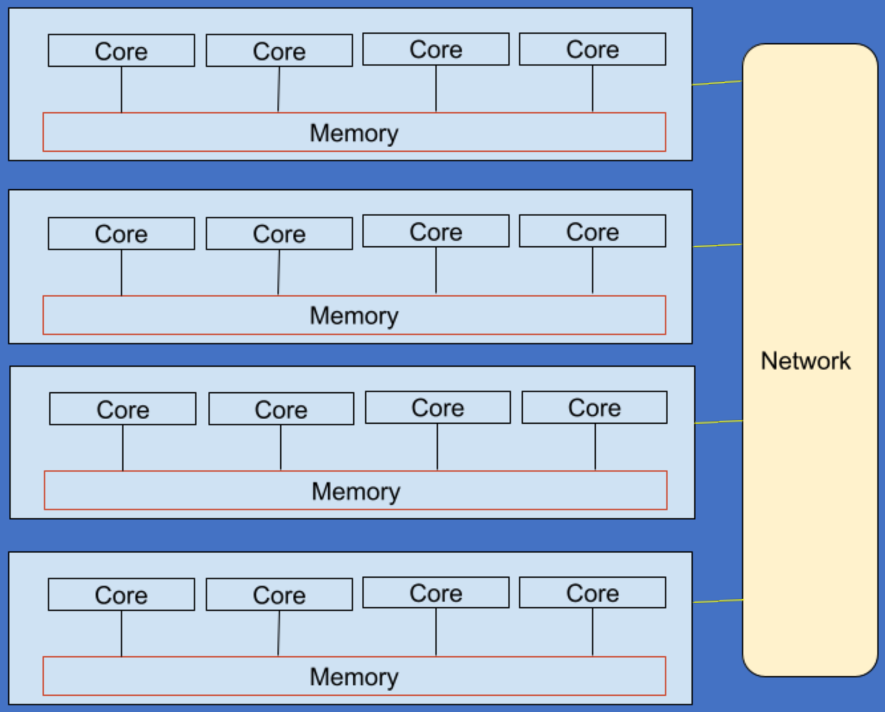
Or GPU card with thousands of cores

Processes have
Independent Memory spaces,
Must communicate to share data



Single machine

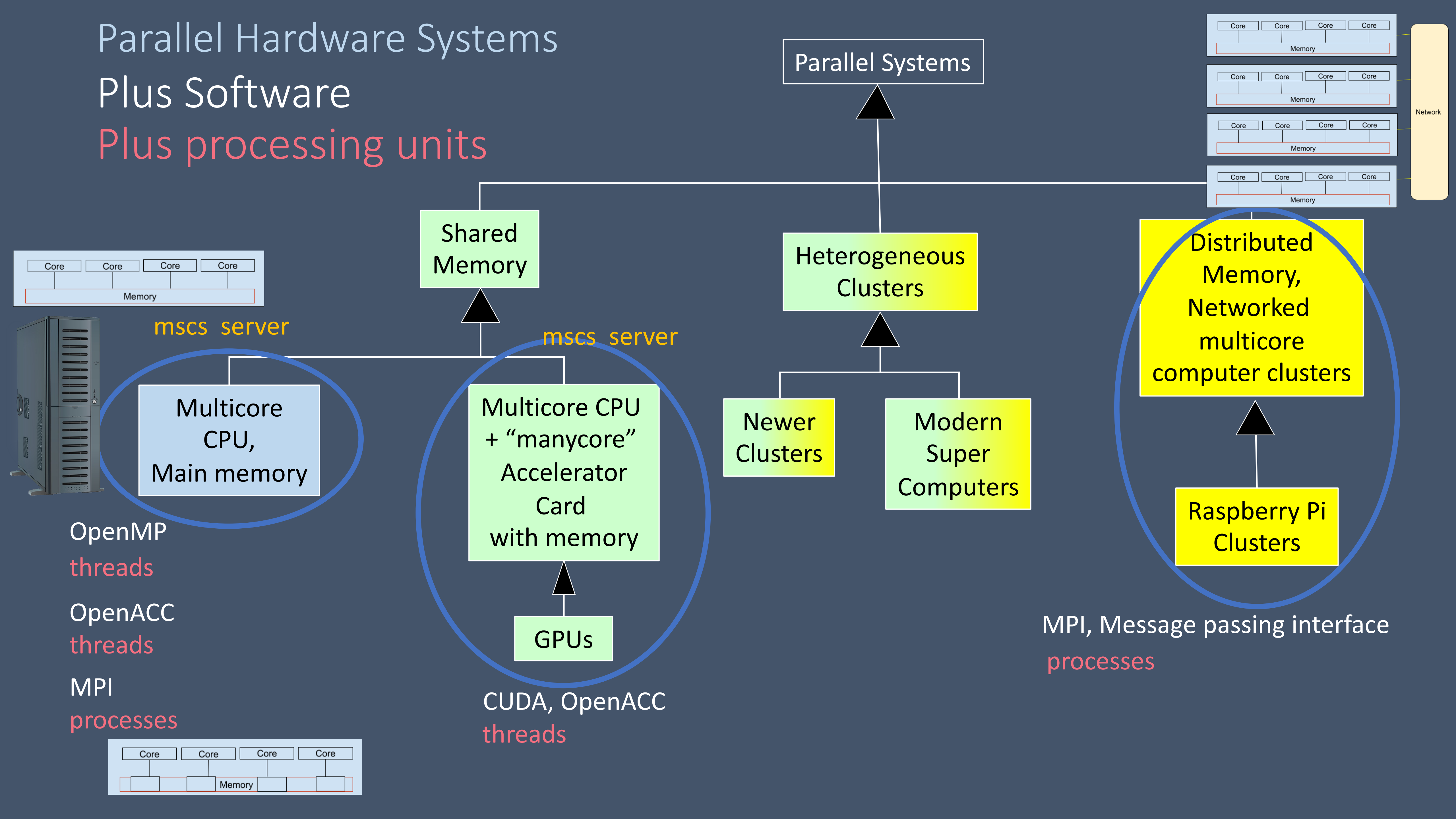
Or



Cluster of machines

Everything is multicore!

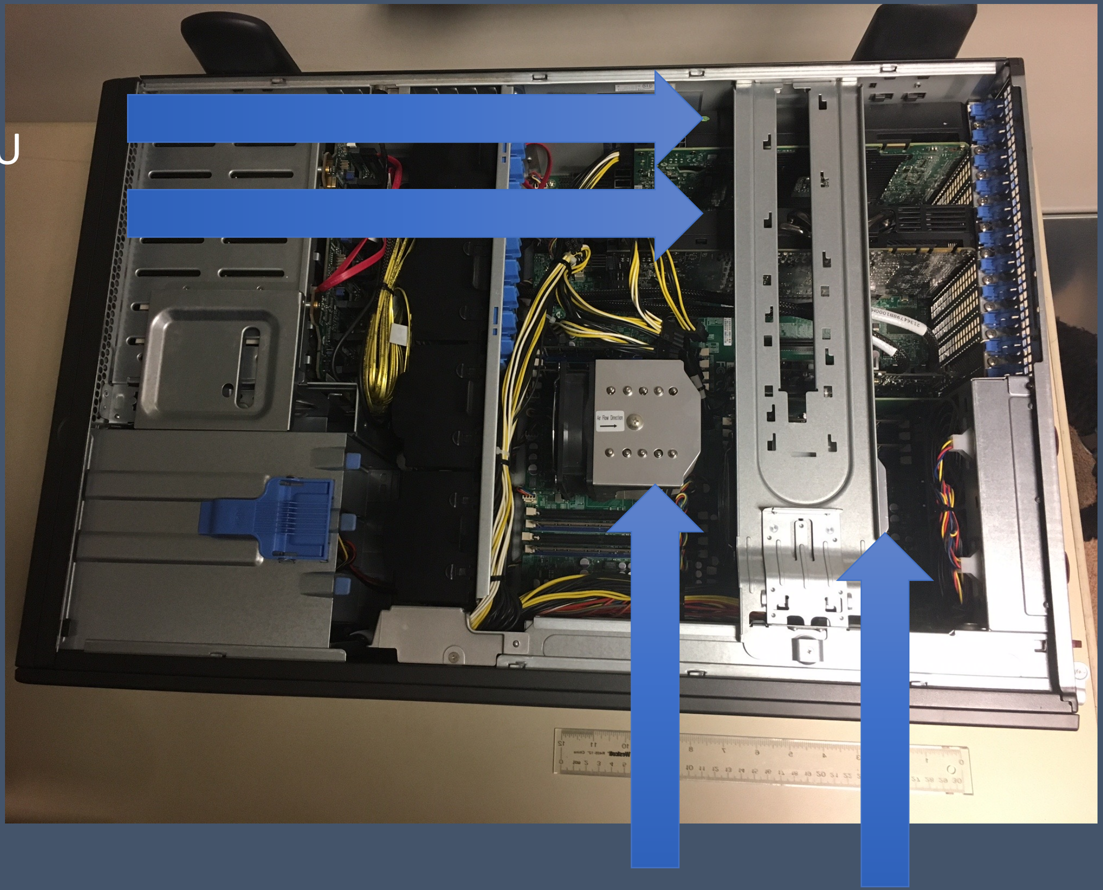
Computing using any combination of these together is referred to as PDC



Parallel Hardware Systems
Plus Software
Plus processing units
Parallel Systems
Shared Memory
Heterogeneous Clusters
Distributed Memory, Networked multicore computer clusters
Core
Memory
Network
mscs server
Multicore CPU, Main memory
mscs server
Multicore CPU + “manycore” Accelerator Card with memory
GPUs
Newer Clusters
Modern Super Computers
Raspberry Pi Clusters
OpenMP
threads
OpenACC
threads
MPI
processes
CUDA, OpenACC
threads
MPI, Message passing interface
processes

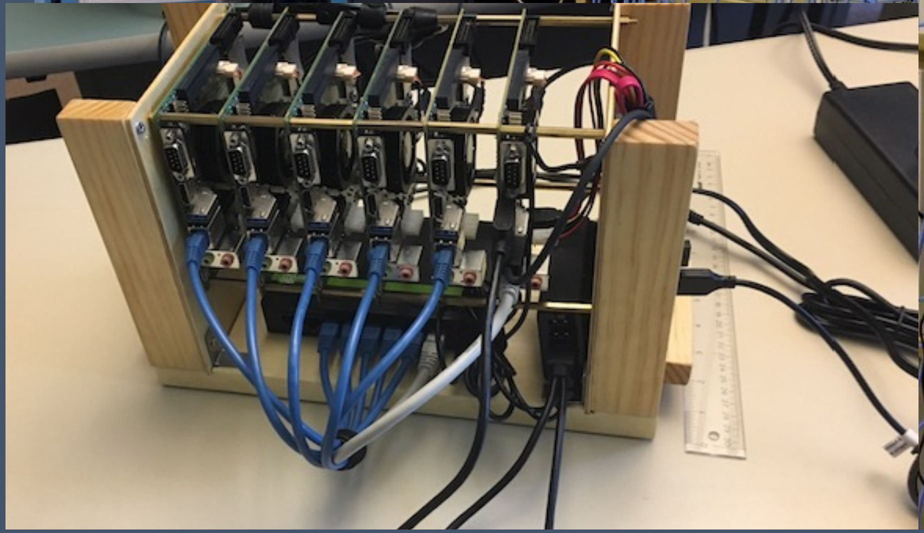
A ‘heterogeneous’ system

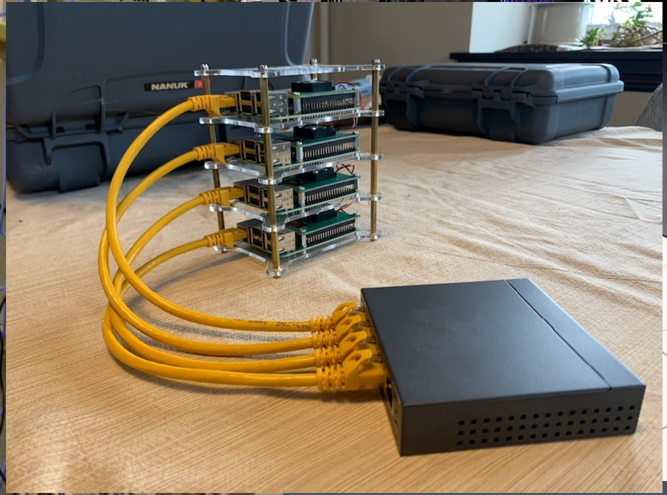
Many-core GPU
cards



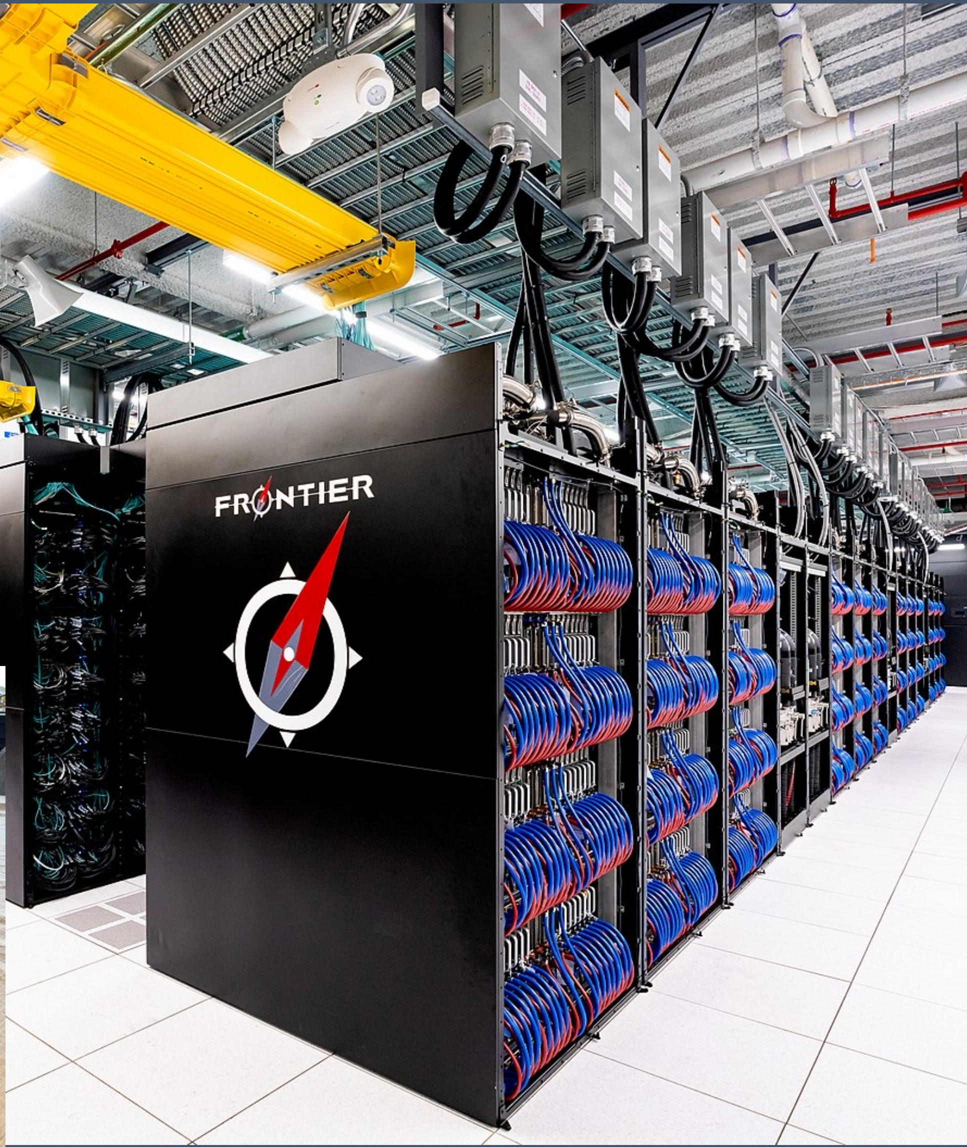
Multicore CPUs
(underneath heat sinks)

Clusters





supercomputers



Beowulf clusters using “COTS”:
Commodity off-the-shelf systems

Course outline

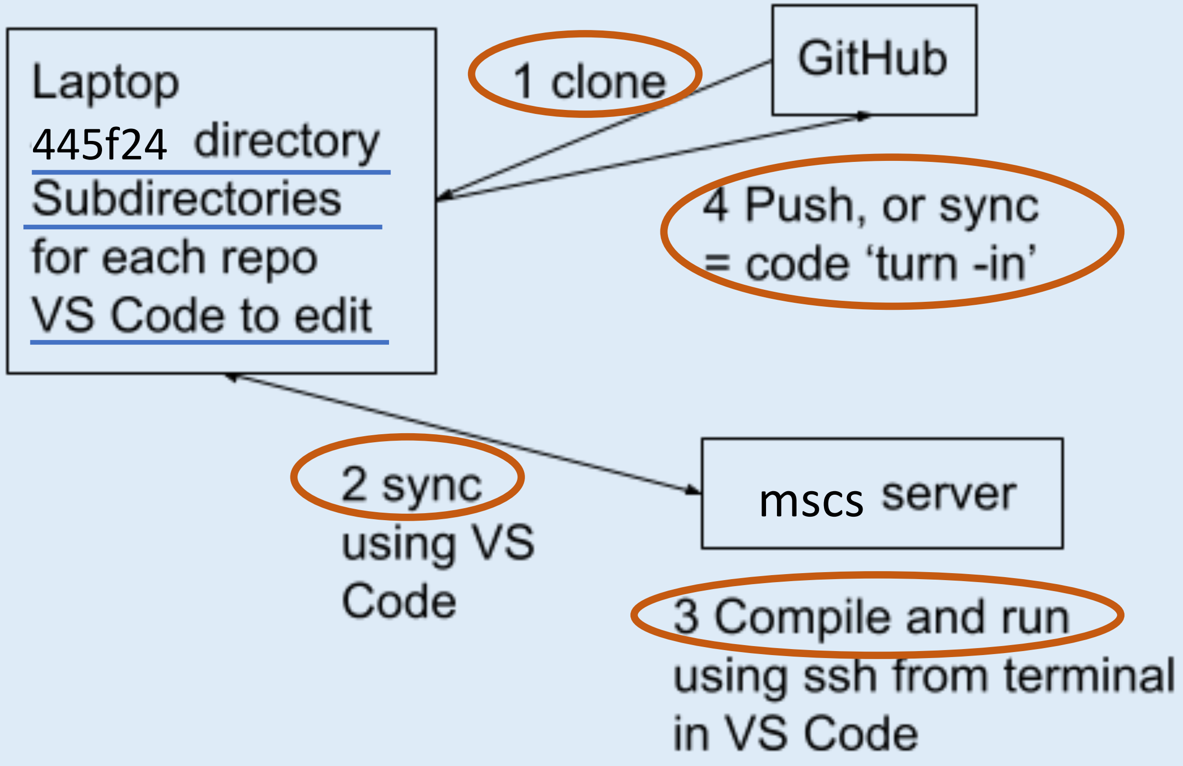
Learn by doing
- Start with some code, not from scratch

Practice Activities, hw0 are low stakes
- Learn and practice as prep for homework
- Suggested reading before class on the schedule

Homework assignments with suggested deadlines (flexible)
- Coding and a report

Course project
- Coding, report, presentation

Method of work on code for course



Reports for homework should be placed on Google Drive provided by instructor